

ForkCast

Review-gated AI policy impact analysis with a Kasper-anchored audit trail.

Submitted for: Conduct Track · Kasper Bounty · GCC Bounty

Live demo: <https://haoyu-haoyu.github.io/ForkCast/>

autonomy

DEMO MODE

DEMO TIME REMAINING
01:14 / 01:30

SCENARIO
London ULEZ Expansion

Checkpoint Control Panel

Human-in-the-loop at every critical step

Overall status
IN PROGRESS

1 Extraction Review

APPROVED

Data sources18 / 18

Stakeholders9

CoverageBlind-safe

Issues0

Open checkpoint

2 Agent Review

PENDING

Agent recommendationProceed

Archetypes36

Policy alignmentHigh

Risks flagged2

Open checkpoint

3 Simulation Replay

PENDING

Simulation window90 days

Events cached128

Key signals6

ModeReplay

Open checkpoint

4 Impact Report

PENDING

Claims reviewed6

Rubric modeBlind

Kasper anchorLive tx

DisclaimersOn

Open checkpoint

Social Feed Simulation Preview

Live preview (simulated)

X policy_advocate

policy_advocate discusses political reputation with stance support.

Support

X car-dependent commuter

car-dependent commuter shows stronger outer-London backlash around cost of compli...

Oppose

X health-conscious cyclist

health-conscious cyclist supports the air-quality rationale while watching imple...

Support

X activist opponent

activist opponent discusses policy reversal with stance oppose.

Oppose

X local councillor

local councillor discusses electoral gain with stance oppose.

Oppose

View full stream

Blind Rubric Coverage

Answer-isolated R1-R6

R1Outer London vs inner London opposition intensity

PARTIAL

R2van drivers / tradespeople / small-business burden

HIT

R3Political salience / electoral risk

HIT

R4Enforcement resistance / camera sabotage narratives

HIT

R5Behavioural adaptation and compliance over time

HIT

R6Benefit/burden balance: air quality vs distributional pain

BALANCED HIT

RUBRIC COVERED

5 / 6

PARTIAL

R1

MECHANISM

Keyword rubric

View detailed rubric report

Impact Summary

Illustrative / demo estimates only

Prediction isolation

No outcome data

Prompt scan

No outcome tokens

Kasper anchoring

Testnet tx live

Report mode

Decision support

Human grading

Pending - see docs/evaluation/ulez_human_adjudication.md

TRUTH STATUS

ULEZ historical case verified

New policies show impact analysis only until a truth_set exists.

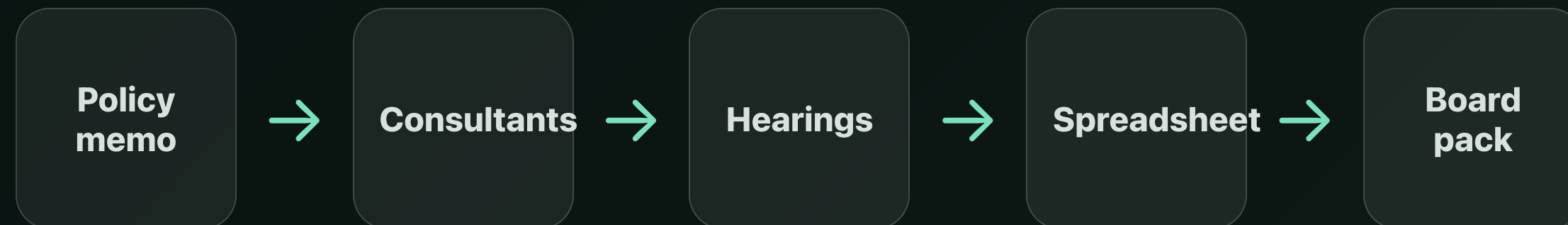
ForkCast

github.com/haoyu-haoyu/ForkCast

THE PROBLEM

Regulatory and policy impact assessment takes weeks.

Enterprise compliance, strategy and public-affairs teams still map stakeholders, model reactions, write assessments and defend assumptions by hand. The reasoning trail often lives in email threads.



Cycle time
3-8 weeks

Traceability
Fragmented

Control
Late review

THE FLOW

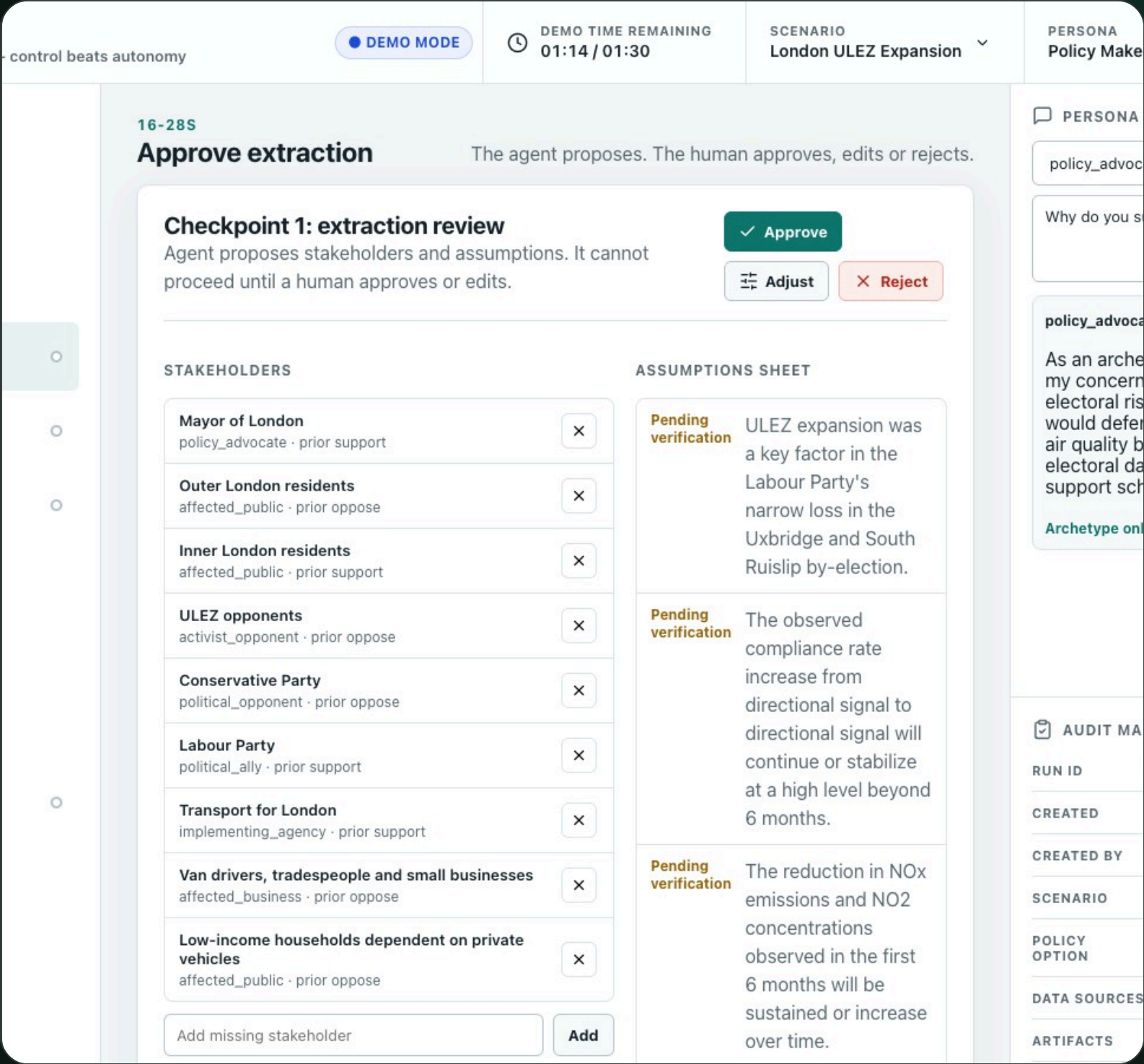
Policy document in. Reviewed, approved, anchored analysis out.

- 1** **Extract** the stakeholder case graph from a policy document.
- 2** **Review and edit** assumptions at a recorded human checkpoint.
- 3** **Run** a replayable impact analysis with provenance labels.
- 4** **Approve the audit trail** and anchor its hash on Kaspia.
- 5** **Fork any reviewed scenario** — conclusions diff side-by-side.

CONTROL BY DESIGN

A fully autonomous agent is a non-starter in an enterprise. We built the opposite.

- ✓ **See everything:** states, evidence, prompts and artifacts stay inspectable.
- ✓ **Approve step-by-step:** weights are editable and diffs are recorded.
- ✓ **Hard limits:** no automatic chain action, no automatic weight changes, no real-person PII.



EVIDENCE DESIGN

Would it have been right? We tested against history with the answers locked away.

Case: London ULEZ expansion 2023. The prediction step records leakage guards and never reads outcome data.

Graded by human adjudication against a source-backed truth set — the published sheet pairs each blind prediction with TfL/YouGov-sourced facts (docs/evaluation/ulez_human_adjudication.md); automated rubric scores are reported as signal coverage only.

Blind prediction

40-46sRun controls

46-52sSimulation replay

52-66sImpact report

66-80sBlind rubric

80-88sAudit + Kaspa

88-90sClose

RULE	RUBRIC ITEM	BLIND PREDICTION SIGNAL	REAL OUTCOME	VERDICT
R1	Outer London vs inner London opposition intensity	Blind prediction identifies outer-London opposition but inner-London contrast is weak.	YouGov/ITV polling (2023-08-09 to 2023-08-14) found 47% of Londoners supported the expansion and 42% opposed; inner London 62% support vs 26% oppose; outer London 51% oppose vs 38% support.	PARTIAL
R2	van drivers / tradespeople / small-business burden	Blind prediction singles out van drivers, tradespeople or small businesses as high-opposition/high-impact groups.	The London-wide ULEZ six month report shows the London-wide compliance rate of subject vehicles rose from 91.6% in June 2023 to 96.2% after six months. Cars reached 97.1%, vans 88.9%.	HIT
R3	Political salience / electoral risk	Blind prediction flags political salience/electoral risk without claiming a specific election result.	The 2023-07-20 Uxbridge and South Ruislip by-election was a Conservative hold: Steve Tuckwell 13,965 votes (45.2%), Danny Beales 13,470 votes (43.6%), majority 495, turnout 46.1%. Credible analysis/reporting linked th...	HIT
R4	Enforcement resistance / camera sabotage narratives	Blind prediction flags enforcement resistance, sabotage, camera backlash or vandalism risk.	A Metropolitan Police FOI response records 416 ULEZ-related offences between 2023-08-01 and 2023-10-15: 59 theft, 357 arson and criminal damage; property status shows 74 stolen and 358 damaged. Sky/ITV reporting notes s...	HIT
R5	Behavioural adaptation and compliance over time	Blind prediction flags short-term backlash with longer-term adaptation/compliance improvement.	The London-wide ULEZ six month report shows the London-wide compliance rate of subject vehicles rose from 91.6% in June 2023 to 96.2% after six months. Cars reached 97.1%, vans 88.9%. Comparing February 2024 with June...	HIT
R6	Benefit/burden balance: air quality vs distributional pain	Blind prediction captures both air-quality/health benefits and distributional fairness.	The six month report covers NOx emissions and NO2 concentrations: outer London car and van NOx emissions were 13% and 7% lower respectively than the no-ULEZ scenario.	BALANCED HIT

electoral damage through targeted support schemes..

Archetype only - no real-person PII

AUDIT MANIFEST

RUN IDulez_2023_phase2_deepseek

CREATED2026-07-01

CREATED BYPolicy Maker (PM)

SCENARIOLondon ULEZ Expansion

POLICY OPTIONULEZ Expansion to Outer London

DATA SOURCES18

ARTIFACTS8

STATUSAnchored on TN-10

KASPA ANCHORINGf553f7bfd73b1ed91b...

Hard limits

No automatic chain action

No automatic weight changes

No real-person PII

Archetypes only

Decision support only

Checkpoint history

Answer isolation first; grading is source-backed.

RED-TEAMED EVALUATION

Our automated scorer looked great —
0.83.
So we attacked it.

Real truth set

0.83

Expected signals were covered.

Inverted truth set

0.83

Verdicts did not move.

Shuffled ×20

0.83

Keyword checklist, not semantic truth.

≠ accuracy

We published the finding, reframed rubric numbers as signal coverage, and moved accuracy evidence to human adjudication.

Evaluation discipline

The metric survived as coverage, not accuracy.

KASPA AUDIT TRAIL

Every run ends in a hash chain.
The head is committed to
Kaspa testnet-10.

- h0

policy input
- h1

extraction
- h2

human approval event
- h3

simulation outputs
- h4

report

New: in-app one-click chain verification returns Overall: PASS against the live chain.

Two real TN-10 anchors: ULEZ f553...3123 · APS 8a68...56b8
FULL TRANSACTION IDS: SLIDE 10

DEMO MODE

DEMO TIME REMAINING
01:14 / 01:30

SCENARIO
London ULEZ Expansion

PERSONA
Policy Maker

kaspa_anchor.json payload.

PASS txid matches anchor
Provided txid matches local anchor txid when one is recorded.

PASS manifest hash matches anchor
Canonical audit_manifest.json hash matches local anchor package.

PASS payload hash matches anchor
Canonical transaction payload hash matches local anchor package.

PASS manifest hash matches tx payload
Canonical audit manifest hash matches on-chain payload commitment.

PASS legacy payload mode
Legacy payload uses manifest-hash commitment mode.

PASS case_graph · case graph

PASS simulation_config · simulation config

PASS agents · agents

PASS blind_prediction · blind prediction

PASS simulation_events · simulation events

PASS impact_report · impact report

PASS backtest_result · backtest result

PASS persona_chat_sample · persona chat sample

OVERALL: PASS

```
{
  "artifact_count": 8,
  "canonicalization": "json.dumps(sort_keys=True, separators=(',', ':'), ensure_ascii=False)",
  "case_id": "ulez_2023_expansion",
  "chain_policy": {
    "decision_support_not_forecast": true,
    "off_chain_ai_reasoning": true,
    "on_chain_payload": "manifest hash commitment only"
  },
  "created_at": "2026-07-01T12:00:00Z"
}
```

Approve anchored payload review

Refuse on-chain anchor

PENDING

POLICY OPTION
ULEZ Expansion to Outer London

SIMULATION WINDOW
90 Days

POPULATION
8.9M

DATA FRESHNESS
Cached 2026-07-01

MODEL VERSION
v0.3.0

NOT A SCRIPTED DEMO

A live GOV.UK consultation went through the pipeline and was anchored.

Input: Automated Passenger Services permitting consultation. For the anchored APS showcase run, a human edited a stakeholder weight, approved it, and broadcast the anchor. New policies without a truth set get impact analysis only — no fake backtests.

What-if forking: compare £12.50 vs £15.00 from the same reviewed baseline; changed conclusions are highlighted side by side.

Fork view

Inner London residents
affected_public · prior support

my

ULEZ opponents
activist_opponent · prior oppose

Conservative Party
political_opponent · prior oppose

Labour Party
political_ally · prior support

Transport for London
implementing_agency · prior support

Van drivers, tradespeople and small businesses
affected_business · prior oppose

Low-income households dependent on private vehicles
affected_public · prior oppose

Add missing stakeholder

Add

DEMO MODE

DEMO TIME REMAINING
01:14 / 01:30

SCENARIO
London ULEZ Expansi

months.

Pending verification

The reduction in NOx emissions and NO2 concentrations observed in the first 6 months will be sustained or increase over time.

Pending verification

Public opinion distribution (directional signal support, directional signal oppose London-wide) remains stable over the simulation period.

Pending verification

Camera vandalism and enforcement resistance will escalate if opposition remains high.

SCENARIO FORKS

Parent: ulez_2023_phase2_deepseek · baseline variant: charge £12.50

Close

SIDE-BY-SIDE COMPARISON

A charge £12.50 · B charge £15.00

21 CHANGED

2 NEW

0 REMOVED

0 UNCHANGED

Illustrative rerun — decision support, not deterministic forecast.

Cached showcase comparison (charge £12.50 vs £15.00) — the live fork API is not reachable in this static demo.

RISK TIMELINE

immediate (0-6 months)

A

stage immediate (0-6 months)

signal Enforcement backlash: 416 ULEZ-related offences including camera vandalism; radical subgroup active.

risk_level high

B

stage immediate (0-6 months)

signal Enforcement backlash: 416 ULEZ-related offences including camera vandalism

risk_level high

long-term (2+ years)

A

stage long-term (2+ years)

signal If compliance stabilizes and air quality improves, political risk may decline; but distributional burden on low-income groups may persist.

risk_level medium

B

stage long-term (2+ years)

signal Risk of sustained radical opposition if distributional burden is not addressed; potential for policy reversal under political pressure

risk_level medium

Side-by-side diffs: risk timeline, claims, stakeholder pressure.

HONEST LIMITS AND ROADMAP

What we claim is exactly what we can prove.

Today

ULEZ evidence path, APS live anchor, in-app verify, scenario forks, and source-backed documentation.

Shipped early

What-if forking was roadmap item B6 — shipped before the deadline.

Next

Truth-set workbench, semantic scorer, chunked ingestion, calibrated magnitude and timing predictions.

ForkCast's moat is evaluation discipline: every new case can add source-backed truth, human adjudication, and a replayable audit trail.

VERIFY IT YOURSELF

Clone the repo. Run the verifier. Check the live demo.

```
git clone https://github.com/haoyu-haoyu/ForkCast && cd ForkCast

# Legacy ULEZ anchor
uv run python scripts/verify_run.py --run-dir runs/ulez_2023_phase2_deepseek --txid
f553f7bfd73b1ed81bd1fd71dbd43631b49c392e20699e82ba2cbc5b263b5123 --network testnet-10

# Showcase APS anchor
uv run python scripts/verify_run.py --run-dir runs/policy_run_20260702T220219Z --txid
8a682481e37f9ca5f006150746e3d3eab003b582621b763f0065885c98c656b8 --network testnet-10
```

Live demo: <https://haoyu-haoyu.github.io/ForkCast/>

Repo: <https://github.com/haoyu-haoyu/ForkCast>